TEXORA · LECTURE SHEET

The Ring Frame

The machine that still spins most of the world's yarn: roving is drafted to its final fineness, twisted by a traveller racing around a ring at up to 40 metres per second, and wound as a cop — simultaneously, on up to two thousand spindles in one frame.

TE-2201 · SPINNING TECHNOLOGY-II · TOPIC 4 · LEVEL 2 TERM 2

1. Principles and objects

The ring frame performs the three final operations of spinning in one pass: **drafting** the roving to the required yarn count (draft about 10–60, up to ~80 with modern high-draft systems), **twisting** the drafted strand to give the yarn its strength, and **winding** the yarn onto a bobbin (the **cop**) in a package shaped for unwinding at the winding machine. Its principle is the ring-and-traveller system: the spindle rotates the cop at full speed; the yarn, passing through a small C-shaped **traveller** clipped on a fixed **ring**, is dragged around the ring by the spindle. Air resistance and ring friction hold the traveller back, so it lags the spindle slightly. That lag does both jobs at once: the rotation of the yarn leg between front roller and traveller inserts **twist**, and the speed difference winds yarn onto the cop at exactly the delivery rate. Modern frames run 1,000–2,000 spindles at spindle speeds of 12,000–25,000 rpm.

2. Main working parts

PART	FUNCTION
Creel	Holds the roving bobbins and lets them unwind freely, without stretching the roving
Drafting system	Three lines of rollers with aprons (Section 3) attenuate roving to yarn count
Lappet (yarn guide)	Centres the yarn over the spindle, controlling the balloon
Spindle and cop	Rotates the package; the winding element (Section 4)
Ring and traveller	Twist insertion and winding control (Section 5)
Ring rail	Traverses slowly up and down, distributing the coils along the cop (Section 6)
Separator plates	Thin blades between spindles that stop neighbouring balloons touching
Tin roller and tapes	The long shaft along the frame that drives every spindle through its tape and wharve

Suction (pneumafil)	Air suction at the drafting zone removes broken-end waste and keeps rollers clean

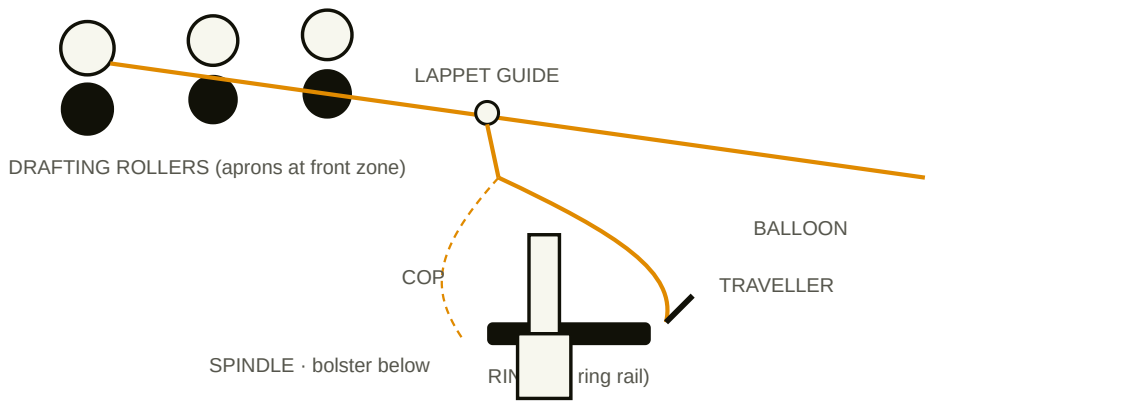


Figure 1 — The yarn path at one spindle: front rollers deliver, the lappet centres the strand, the yarn balloons out to the traveller on the ring, and is wound onto the cop.

3. Drafting system and pressure in the drafting zone

The standard is the **double-apron system**: three lines of bottom rollers; the back zone gives a break draft of about 1.1–1.4; the front (main) zone drafts between the middle rollers — whose top roller works through a pair of aprons held in a cradle — and the front rollers. The aprons deliver the fibres to within a few millimetres of the front nip, controlling floating fibres at drafts that the roving frame never attempts.

Applying pressure (weighting). A drafting nip only works if the top rollers are pressed onto the strand hard enough to grip fibres without crushing them. Three generations of weighting coexist:

- **Dead-weight and lever systems** — hanging weights load each top roller through levers; simple, but the load varies with position and maintenance.
- **Spring-loaded arms** — a pivoting arm with calibrated springs presses all three top rollers as one unit, quick to lift for piecing-up.
- **Pneumatic loading (modern standard)** — an air-pressurised arm applies an adjustable, uniform load (front-roller loads of roughly 15–25 daN per spindle position are typical) that stays constant over the frame's life and can be tuned per material.

The top rollers themselves are covered with synthetic rubber **cots** (Section 5.2). Pressure, cot hardness and roller setting act together: insufficient pressure lets fibres slip in bundles (uneven, slubby yarn and high end-breaks); excessive pressure damages fibres, laps the rollers and overloads the drives.

4. Parts of a spindle

The spindle is a precision component expected to run true at 25,000 rpm for years. From the top: the **blade** — the slim steel rod that carries the bobbin, tapered so the cop tube fits by friction; the **wharve** — the small pulley on the blade, driven by the tape from the tin roller; and the **bolster** — the fixed bearing housing in which the blade's footstep rotates, containing a spiral-grooved bearing surface and an oil reservoir that continuously lubricates the footstep and damps vibration. A spindle brake (a small knee- or hand-operated stop on the bolster) halts one spindle for piecing-up while the frame runs. Spindle gauge (spacing) and blade length set how many spindles fit a frame and how large a cop can be built.

5. Ring and traveller; apron and cot

5.1 Ring and travellers, and the ring data system

The ring is a hardened, polished steel circle fixed in the ring rail; common diameters are 36–51 mm (finer counts and higher speeds use smaller rings). Its cross-section has a **flange** — the rim on which the traveller runs — specified by **flange width**: the radial width of that running rim, supplied in sizes such as 3.2, 3.8, 4.5 and 5.1 mm. The traveller must match the flange: too narrow a flange for the traveller and it rocks and burns; too wide and the traveller runs loosely and yarn tension control is lost. Travellers are graded by weight (finer yarn → lighter traveller) and by shape (ear, bow and hook profiles trade yarn clearance against stability).

Choice is governed by the **ring data** for the job — the matched set of ring diameter, flange width, traveller weight and spindle speed for each yarn count, built up from the frame maker's tables and the mill's own trials. Working limits that these tables respect:

YARN COUNT (NE)	RING DIAMETER (MM)	TRAVELLER SPEED LIMIT	TYPICAL SPINDLE SPEED (RPM)
Coarse, 10–20	45–51	Traveller linear speed ≈ 35–45 m/s maximum; beyond it the traveller overheats, wears and flies off	10,000–13,000
Medium, 20–40	38–45		13,000–18,000
Fine, 40–80+	36–38		18,000–25,000

Modern frames record and display this data continuously — spindle speed, delivery, twist, production and end breaks per position group — as the machine's **data system**, used by the mill to hold settings true across shifts. The principle behind the limit is physical: traveller heating rises with the cube of

rubbing speed and with traveller mass, so fine yarns (light travellers) can use high spindle speeds only on small rings.

5.2 Apron and cot roller

The **aprons** are endless flexible bands (leather originally, now synthetic rubber with fabric reinforcement) looped over the middle bottom roller and a guide bar; the top apron is carried in a cradle that sets the gap between the apron noses — the actual fibre-control length. The **cot roller** is any top roller: a steel core covered with a synthetic rubber cot, ground to an exact diameter and chosen by hardness (about 65°–85° Shore A; softer cots grip fibres more gently and suit fine counts and man-made fibres, harder cots resist wear and grooving). Buffing and re-grinding cots on schedule is essential: a grooved cot prints its defect into every metre of yarn.

6. Winding and the building mechanism

The ring rail traverses slowly; its motion has two components — a short, quick traverse that lays the coils (the **wind**) and a slow rise that shifts the whole traverse upward so layers accumulate (the **bind**), combined by the builder motion (ratchet, chain and cam or servo) into the cop's shape: a conical tip, a full body — the chase — and a slightly tapered base. The build matters commercially: a well-built cop holds the maximum yarn, never sloughs coils in handling, and unwinds over-end at winding-machine speeds without snapping. Doffing — full cops off, empty tubes on — is manual on simple frames and automatic, with cop transport to the winder, on modern ones.

7. Settings, speeds and production

Principal change points: draft gears (count), twist gear, spindle speed (motor pulley or inverter), ring rail builder settings, and traveller weight. Twist follows the same multiplier law as all cotton spinning:

$$\text{TPI} = \text{TM} \times \sqrt{(\text{Ne})} \quad \cdot \quad \text{Delivery (in/min)} = \text{Spindle rpm} \div \text{TPI}$$

WORKED EXAMPLE — RING FRAME PRODUCTION PER SPINDLE

Yarn 30 Ne, TM = 3.6, spindle speed 15,000 rpm, efficiency 95 %.

TPI = 3.6 × √30 = 3.6 × 5.48 ≈ **19.7 TPI**.

Delivery = 15,000 ÷ 19.7 ≈ 761 in/min ≈ 21.1 yd/min ≈ 1,268 yd/h per spindle.

One pound of 30 Ne contains 30 × 840 = 25,200 yd, so production = 1,268 ÷ 25,200 ≈ 0.0503 lb/h ≈ **22.8 g/h per spindle**.

A 1,000-spindle frame at 95 % efficiency therefore delivers ≈ 22.8 kg/h × 0.95 ≈ **21.7 kg/h** of 30 Ne yarn.

8. End breakage rate

The **end breakage rate** — breaks per 100 spindle-hours — is the ring frame's health indicator: good running on medium counts is in single figures; a rising rate stops the frame being economical long before it stops being possible (an operative can piece up only so many ends). Breaks occur where yarn tension peaks exceed the strength of the weakest place in the strand. Causes group into: **material** (short fibre, under-twisted roving, uneven roving), **drafting** (worn cots, wrong settings, damaged aprons), **twisting/winding** (worn or wrong travellers, grooved rings, excessive spindle speed or balloon tension, misaligned lappets), and **housekeeping** (fly accumulation, damaged spindle tapes). The remedy list is read in the same order — most "machine problems" begin in the roving.

9. Doubling — making plied yarns

Two or more single yarns twisted together give **plied (folded) yarn** — stronger, rounder, more even, and essential for sewing threads, tyre cord, quality knitting yarns and crepes. The systems:

SYSTEM	PRINCIPLE	CHARACTER
Ring doubling frame	Like a ring frame without drafting: doubled yarns pass through a water trough (wet doubling for smooth threads) or dry, and are twisted by ring and traveller	Good twist control; slow; classic for sewing threads
Two-for-one twister (TFO)	Yarn from a stationary package passes down through the hollow spindle and balloons around it: one spindle revolution inserts **two** twists	Double the twist per rpm of a ring doubler; large packages, no traveller wear; the modern standard for filament and spun ply yarns
Up-twister / direct cabling	Two yarns are combined and twisted around each other in one pass from supply packages	Used for cords and carpets; exact length balance between components
Doubling on the winder (assembly winding)	Not twisting at all — two ends wound together onto one package as preparation for the twister	Guarantees equal lengths entering the ply, preventing one tight and one slack component

Glossary

Cop — the yarn package built on the ring-frame bobbin. **Traveller** — the small C-clip on the ring that the yarn drags around; its lag inserts twist and winds the yarn. **Balloon** — the rotating curve of yarn between lappet and traveller. **Wharve** — the spindle's drive pulley. **Bolster** — the spindle's bearing housing with its oil reservoir. **Cot** — the rubber covering of a top drafting roller. **Flange width** — the radial width of the ring rim on which the traveller runs. **Builder motion** — the mechanism producing

the cop build by combining traverse and lift. **End breakage rate** — broken ends per 100 spindle-hours. **TFO** — two-for-one twister, inserting two twists per spindle revolution.

Key points

• Twist and winding come from one fact: the traveller lags the spindle; the rotation twists, the lag winds.
• Drafting is 3-roller double-apron; pressure is applied by dead weight, springs or — modern standard — pneumatics, acting through rubber cots.
• Ring diameter, flange width, traveller weight and spindle speed form one matched data set; traveller speed (~35–45 m/s) is the hard limit.
• The builder motion combines coil traverse and layer lift to form the cop.
• End breakage rate (per 100 spindle-hours) is the frame's health gauge; most causes start in roving quality or worn drafting parts.
• Doubling: ring doubling (classic), TFO (two twists per revolution, modern standard), up-twisting and assembly winding as preparation.

Practice

1. A frame spins 20 Ne with TM 4.0 at 12,000 rpm. Calculate the TPI, delivery in yd/h per spindle, and production in g/h per spindle.
2. A traveller runs on a 45 mm ring at 18,000 rpm spindle speed. Calculate the traveller's approximate linear speed (m/s) and judge whether it is within the working limit.
3. Explain why raising spindle speed raises production less than proportionally once end breaks are considered.
4. A mill changes from 40 Ne to 60 Ne on the same frames. List four changes the ring data for the job requires.
5. Why must the traveller be *lighter* for finer yarn, even though finer yarn is weaker and "easier to pull around"?
6. Compare ring doubling and two-for-one twisting on: twists per spindle revolution, package size, traveller wear, and typical applications.

Answer key

1. TPI = 4.0 × √20 ≈ 4.0 × 4.47 ≈ **17.9 TPI**. Delivery = 12,000 ÷ 17.9 ≈ 671 in/min ≈ 18.6 yd/min ≈ **1,118 yd/h**. One pound of 20 Ne = 16,800 yd, so production = 1,118 ÷ 16,800 ≈ 0.0665 lb/h ≈ **30.2 g/h per spindle**.

2. Circumference = π × 0.045 m ≈ 0.141 m. Speed ≈ 0.141 × 18,000 ÷ 60 ≈ **42.4 m/s** — at the top of the 35–45 m/s working range: permissible only with a light traveller in good condition, and a further speed increase would overheat and wear it rapidly.
3. Higher spindle speed raises balloon tension and the tension peaks at weak places, so the end breakage rate climbs; each break stops that spindle's production until pieced up and costs operative time, so effective efficiency falls as speed rises. Beyond the optimum, extra rpm buys more stoppage than output.
4. (i) Smaller ring diameter (toward 36–38 mm) or at least re-check traveller speed; (ii) lighter traveller weight matched to the finer yarn; (iii) higher twist (TM × √60) and corresponding twist gear change, with spindle speed re-set; (iv) higher draft change gears and finer roving hank verified; also lighter pneumatic weighting/softer cots may be set for the finer strand (any four consistent changes).
5. Yarn tension during spinning is generated largely by the traveller's centrifugal force and friction, both rising with traveller weight. Fine yarn has little strength to spare, so the tension budget must be cut — a lighter traveller produces less centrifugal pull and less balloon tension, keeping tension peaks below the yarn's breaking load. The traveller is matched to the yarn's strength, not its convenience.
6. Ring doubling: one twist per revolution (like the ring frame), small cop packages, traveller wear and speed limit apply; classic for sewing threads including wet doubling. TFO: two twists per revolution, large stationary supply packages and large delivery packages, no traveller at all (balloon around the package), the standard for high-production plying of spun and filament yarns.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib